

Ex. 4. Simplify : $\frac{112}{\sqrt{196}} \times \frac{\sqrt{576}}{12} \times \frac{\sqrt{256}}{8}$.

Sol. Given expression = $\frac{112}{14} \times \frac{24}{12} \times \frac{16}{8} = 8 \times 2 \times 2 = 32$.

Ex. 5. If $a * b * c = \frac{\sqrt{(a+2)(b+3)}}{c+1}$, then find the value of $6 * 15 * 3$.

Sol. $6 * 15 * 3 = \frac{\sqrt{(6+2)(15+3)}}{3+1} = \frac{\sqrt{8 \times 18}}{4} = \frac{\sqrt{144}}{4} = \frac{12}{4} = 3$.

Ex. 6. Find the value of $\sqrt{1\frac{9}{16}}$.

Sol. $\sqrt{1\frac{9}{16}} = \sqrt{\frac{25}{16}} = \frac{\sqrt{25}}{\sqrt{16}} = \frac{5}{4} = 1\frac{1}{4}$.

Ex. 7. What is the square root of 0.0009?

Sol. $\sqrt{0.0009} = \sqrt{\frac{9}{10000}} = \frac{\sqrt{9}}{\sqrt{10000}} = \frac{3}{100} = 0.03$.

Ex. 8. Evaluate $\sqrt{175.2976}$.

Sol. Method: We make even number of decimal places by affixing a zero, if necessary. Now, we mark off periods and extract the square root as shown.

$$\therefore \sqrt{175.2976} = 13.24$$

Ex. 9. What will come in place of question mark in each of the following questions?

$$(i) \sqrt{\frac{32.4}{?}} = 2 \quad (ii) \sqrt{86.49} + \sqrt{5 + (?)^2} = 12.3$$

Sol. (i) Let $\sqrt{\frac{32.4}{x}} = 2$. Then, $\frac{32.4}{x} = 4 \Leftrightarrow 4x = 32.4 \Leftrightarrow x = 8.1$.

$$(ii) \text{ Let } \sqrt{86.49} + \sqrt{5 + x^2} = 12.3.$$

$$\text{Then, } 9.3 + \sqrt{5 + x^2} = 12.3 \Leftrightarrow \sqrt{5 + x^2} = 12.3 - 9.3 = 3$$

$$\Leftrightarrow 5 + x^2 = 9 \Leftrightarrow x^2 = 9 - 5 = 4 \Leftrightarrow x = \sqrt{4} = 2.$$

Ex. 10. Find the value of $\sqrt{\frac{0.289}{0.00121}}$.

(IGNOU, 2003)

Sol. $\sqrt{\frac{0.289}{0.00121}} = \sqrt{\frac{0.28900}{0.00121}} = \sqrt{\frac{28900}{121}} = \frac{170}{11}$.

Ex. 11. If $\sqrt{841} = 29$, then find the value of $\sqrt{841} + \sqrt{8.41} + \sqrt{0.0841} + \sqrt{0.000841}$.

(M.B.A., 2006)

Sol. Given expression = $\sqrt{841} + \sqrt{\frac{841}{10^2}} + \sqrt{\frac{841}{10^4}} + \sqrt{\frac{841}{10^6}} = \sqrt{841} + \frac{\sqrt{841}}{10} + \frac{\sqrt{841}}{10^2} + \frac{\sqrt{841}}{10^3}$
 $= 29 + \frac{29}{10} + \frac{29}{100} + \frac{29}{1000} = 29 + 2.9 + 0.29 + 0.029 = 32.219$.

Ex. 12. If $\sqrt{1 + \frac{x}{144}} = \frac{13}{12}$, then find the value of x .

1	175.2976 (13.24
	1
23	75
	69
262	629
	524
2644	10576
	10576
	×